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Short-Term Wind Power Forecasting with Gradient Boosting and Numerical Weather Prediction Data

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ABSTRACT

The growing share of wind generation in power systems raises the requirements for the accuracy of its short-term forecasting, on which operational planning and balancing costs depend. This study proposes an approach to forecasting the output of a wind farm over horizons from 1 to 48 hours that combines gradient boosting with features extracted from numerical weather prediction forecasts. The data come from an anonymised wind farm with an installed capacity of 120 MW for 2021–2025 at hourly resolution, together with forecasts of wind speed and direction at several heights for nine grid nodes around the farm. In addition to the raw meteorological variables, features describing the spatial gradient of wind speed, vertical wind shear and atmospheric stability were constructed. The model was compared with persistence, a model based on the manufacturer's power curve and a recurrent neural network. The proposed model reduced the normalised root-mean-square error at the 24-hour horizon to 9.8%, compared with 12.6% for the power-curve model and 10.9% for the neural network. The largest gains are achieved during abrupt changes in wind speed associated with the passage of weather fronts. Feature importance analysis showed that the spatial wind speed gradient contributes substantially to forecast quality at horizons beyond 12 hours. The approach has low computational cost and can be deployed in dispatch practice.

Keywords: wind energy; power forecasting; gradient boosting; numerical weather prediction; time series

Краткосрочный прогноз ветровой генерации с помощью градиентного бустинга и данных численных моделей погоды

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АННОТАЦИЯ

Рост доли ветровой генерации в энергосистемах повышает требования к точности её краткосрочного прогноза, от которой зависят планирование режимов и стоимость балансирования. В работе предложен подход к прогнозу выработки ветровой электростанции на горизонте от 1 до 48 часов, сочетающий градиентный бустинг с признаками, извлечёнными из прогнозов численных моделей погоды. Использованы данные условной ветровой электростанции установленной мощностью 120 МВт за 2021–2025 гг. с часовым разрешением, а также прогнозы скорости и направления ветра на нескольких высотах для девяти узлов сетки вокруг станции. Помимо исходных метеорологических переменных, сформированы признаки пространственного градиента скорости ветра, сдвига ветра по высоте и устойчивости атмосферы. Сравнение проводилось с методом персистентности, моделью на основе паспортной кривой мощности и рекуррентной нейронной сетью. Предложенная модель снизила нормированную среднеквадратичную ошибку на горизонте 24 часа до 9,8 % против 12,6 % у модели на основе кривой мощности и 10,9 % у нейронной сети. Наибольший выигрыш достигается в периоды резких изменений скорости ветра при прохождении атмосферных фронтов. Анализ важности признаков показал, что пространственный градиент скорости ветра вносит существенный вклад в качество прогноза на горизонтах свыше 12 часов. Предложенный подход отличается невысокими вычислительными затратами и может быть внедрён в практику диспетчерских служб.

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Ключевые слова: ветровая энергетика; прогноз генерации; градиентный бустинг; численный прогноз погоды; временные ряды

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1 INTRODUCTION

The output of wind farms is determined by wind speed, which changes rapidly and is poorly predictable. Forecast errors make it necessary to hold capacity in reserve and incur balancing costs, so improving forecast accuracy has a direct economic effect.

Numerical weather prediction forecasts are the main source of information about future wind, but their spatial resolution is insufficient to account for local conditions. The aim of the study is to develop a model that extracts additional information from weather forecasts by means of feature engineering and gradient boosting.

2 MATERIALS AND METHODS

Hourly output data from the 40 wind turbines of an anonymised 120 MW farm for January 2021 to June 2025 were used, together with an archive of numerical weather prediction forecasts with lead times of up to 48 hours for nine grid nodes. A total of 86 features were constructed, including wind speed at heights of 10, 100 and 150 m, wind shear, the spatial wind speed gradient and an atmospheric stability indicator.

The gradient boosting model was trained separately for each group of forecast horizons. Data for 2021–2023 were used for training, 2024 for hyperparameter tuning and 2025 for the final evaluation. Performance was measured by the normalised root-mean-square error (NRMSE) relative to installed capacity.

3 RESULTS

The dependence of the error on the forecast horizon for the models compared is shown in Fig. 1. At horizons of up to three hours, the persistence method is competitive, but from six hours onwards the models based on weather forecasts substantially outperform it.

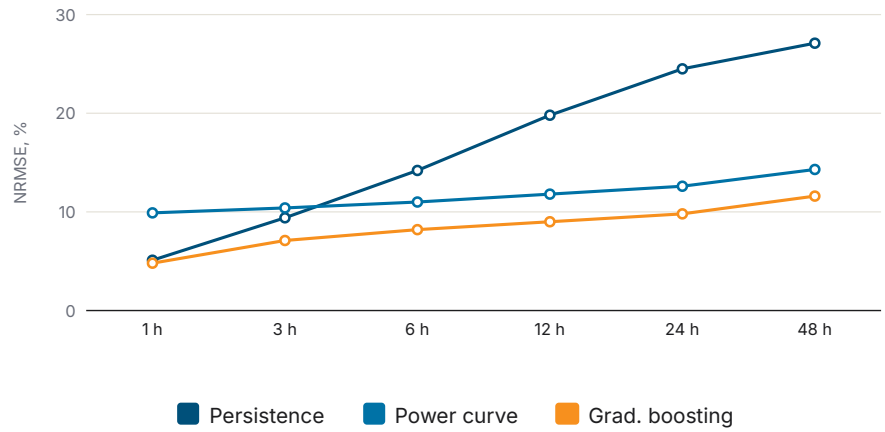


Fig. 1. Normalised root-mean-square forecast error as a function of horizon, %

Summary results for the 24-hour horizon are given in Table 1. The proposed model outperforms all alternatives, and during the passage of weather fronts its advantage over the power-curve model reaches 4.5 percentage points.

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Table 1. Forecast performance at the 24-hour horizon (2025 test period)

Model	NRMSE, %	NMAE, %	Training time
Persistence	24,5	18,2	—
Power curve	12,6	9,1	—
Recurrent network	10,9	7,8	3.5 h
Gradient boosting	9,8	7,0	6 min

4 DISCUSSION

The advantage of the proposed model is explained primarily by the use of spatial features: the wind speed gradient between grid nodes makes it possible to “see” an approaching front earlier. The recurrent network achieves similar performance but requires an order of magnitude more computational resources and is harder to interpret.

A limitation of the study is the use of data from a single farm; the transferability of the features to farms in other climatic conditions requires verification.

5 CONCLUSION

Gradient boosting with features extracted from numerical weather prediction forecasts provides accurate and computationally efficient short-term wind power forecasting suitable for practical application.

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